Ratios & Proportional Reasoning: Unit Rates & Proportions

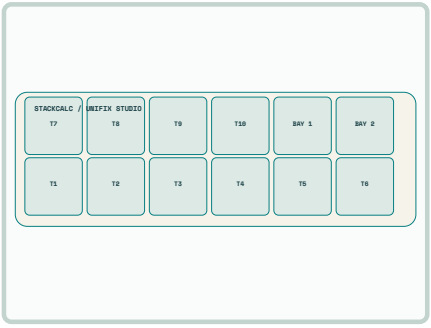
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.7.RP.A.2

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Unifix Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Arrange Counting & Array tiles in repeated ratio groups: 3 teal to 2 amber. Extend to 6:4, 9:6.

Sketch Model on Paper:
Draw a ratio table and coordinate graph showing straight line through origin (0,0).

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1.	180	X: 180	Total miles (180) in
2.	ENTER	Y: 180 | X: 180	Stack lift
3.	6	Y: 180 | X: 6	Gallons (6) into X
4.	÷	X: 30	Unit rate: 180 ÷ 6 =

EXPECTED LCD X: 30 mpg (k = 30)

1 **A store sells 4 notebooks for $16. Find cost per notebook.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **A runner completes 10 laps in 25 minutes. Find minutes per lap.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Solve proportion: 2 / 5 = x / 30. Find x: (2×30) ÷ 5.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Find constant of proportionality: y = 56 when x = 7.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 4

Ratios & Proportional Reasoning: Unit Rates & Proportions • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Counting & Array Studio (COUNT-01) & Ratio Scale

CAD Blueprint: unifix_tray

Target Device: SC-6 / SC-32 Bridge

OpenSCAD Source/ STL Source File:

Education/scad/counting_array_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Case & 4-Color Array Tiles (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**A store sells 4 notebooks for $16. Find cost** Standard: CCSS.MATH.7.RP.A.2	**$4.00/notebook**	[1][6] [ENTER] [4] [÷]
2	**A runner completes 10 laps in 25 minutes. Fi** Standard: CCSS.MATH.7.RP.A.2	**2.5 min/lap**	[2][5] [ENTER] [1][0] [÷]
3	**Solve proportion: 2 / 5 = x / 30. Find x: (2** Standard: CCSS.MATH.7.RP.A.2	**x = 12**	[2] [ENTER] [3][0] [×] [5] [÷]
4	**Find constant of proportionality: y = 56 whe** Standard: CCSS.MATH.7.RP.A.2	**k = 8**	[5][6] [ENTER] [7] [÷]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Watch for inverting the ratio (calculating x/y instead of y/x for unit price).

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.